UNIT 1 · 9 hours

Power and Energy



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
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Learning objectives

- Define electrical power and energy with units
- Use $P = VI$, I^2R and V^2/R
- Convert between joule, watt-hour and kilowatt-hour
- Compute energy consumed and its cost

Syllabus context

Basic nodal and mesh analysis excited by independent DC voltage sources, Power and Energy. Generation of sinusoidal voltage, frequency, average value, RMS value, form factor and peak factor, phasor representation of alternating quantities.

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Power and energy: definitions

Energy (U)

Capacity to do work. Unit: joule (J). Electrical energy: watt-hour (Wh), kilowatt-hour (kWh).

Power (P)

Rate of doing work: P = U / t.

Unit: watt, 1 W = 1 J/s.

Conversions

1 J = 1 W·s

1 Wh = 3600 J

1 kWh = 3.6 MJ = 1 'unit'

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Electrical power formulas

$$P = V I$$

from V = W/Q and I = Q/t

$$P = I^2 R$$

substitute V = I R

$$P = V^2 / R$$

substitute I = V / R

- Power is absorbed by a load (resistor) and delivered by a source.
- I^2R is the heating loss in conductors; it grows with the square of the current.

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Example 1: DC motor rating

Problem A 24 V DC motor has a rated current of 33.5 A. What electrical power does it take?

- $P = V \times I$
- $P = 24 \times 33.5$

Answer: P = 804 W

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Example 2: energy of a lamp

Problem How much energy (kWh) is needed to light a 60 W bulb continuously for 1 year (365 days)?

- Time $t = 365 \times 24 = 8760$ h
- $U = P \times t = 60 \times 8760 = 525\,600$ Wh
- Convert to kWh: divide by 1000

Answer: U = 525.6 kWh

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Example 3: cost of running a heater

Problem A 2 kW heater runs 3 h per day for 30 days. Tariff ₹6 per unit. Find energy and cost.

- Energy = 2 kW × 3 h × 30 = 180 kWh (units)
- Cost = 180 × 6

Answer: 180 kWh, ₹1080

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Power in a source-resistor circuit

- Power delivered by sources = power absorbed by resistors (conservation of energy).
- Check: mesh example, 10 V source gives 19.64 W; 18 Ω absorbs 3.59 W; the rest is absorbed by 1 Ω and 2 Ω and the 5 V source.
- Source power: P = E·I; it can be negative if the current flows into the + terminal (the source is charging).
- Always verify the power balance after solving a network.

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Key takeaways

- $P = VI = I^2R = V^2/R$ (watts)
- Energy = power × time; 1 Wh = 3600 J; 1 kWh = 1 unit
- Power delivered = power absorbed in a circuit
- Heating loss varies as the square of the current

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Practice questions

Q1. A 230 V, 1 kW iron is used for 2 h. Find energy and current.

Q2. Find the resistance of a 100 W, 230 V lamp.

Q3. How many kWh does a 5 W LED bulb use in a year if on for 6 h/day?

Q4. Why does doubling current quadruple the conductor loss?